

Water Segmentation Using U-Net

Project Summary

This project implements a semantic segmentation model to extract water bodies from multispectral satellite images using the U-Net architecture in PyTorch. The workflow included loading and preprocessing the dataset, creating custom Dataset and DataLoader classes, implementing the U-Net model, training it, evaluating its performance, and generating segmentation predictions.

Methodology

The dataset consisted of 306 images (244 for training and 62 for validation). Each image contains 12 spectral bands with a spatial size of 128×128 pixels. The model was trained for 25 epochs using the Adam optimizer and Binary Cross Entropy (BCE) loss with a batch size of 16. Performance was monitored using Dice Score and Intersection over Union (IoU).

Results

The model showed consistent improvement throughout training, with decreasing training and validation losses and increasing Dice and IoU scores. Final results were:

- Training Loss: 0.2342
- Validation Loss: 0.2088
- Dice Score: 0.8547
- IoU Score: 0.7483
- Precision: 0.8554
- Recall: 0.8641
- F1-Score: 0.8597

Conclusion

The implemented U-Net successfully segmented water bodies from multispectral satellite imagery and achieved strong quantitative results. This project demonstrates a complete deep learning pipeline for semantic segmentation and provides a solid foundation for future improvements such as Attention U-Net, DeepLabV3+, and advanced data augmentation techniques.